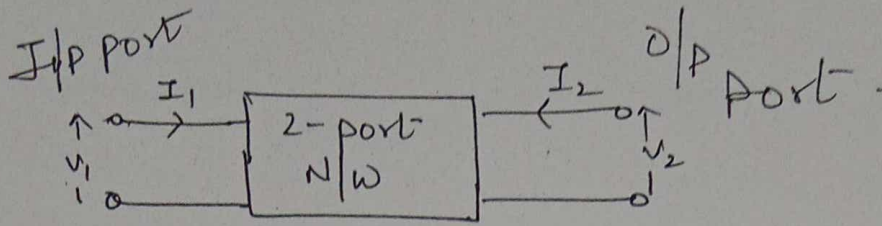


5. Two-port N/w

①-①



∴ I entering 1 terminal of a pair is equal & opposite to the I leaving other terminal of the pair then this type of terminal pair is called as a "port".

2-port N/w - represented by 4 variables.
2 voltages (V_1, V_2) & 2 currents (I_1, I_2)

Relationship of 2 port Variables :

no. of possible combi generated by 4 variables taken 2 at a time = 6. So total 6 combi are there with which we can analyse any type of 2-port N/w.

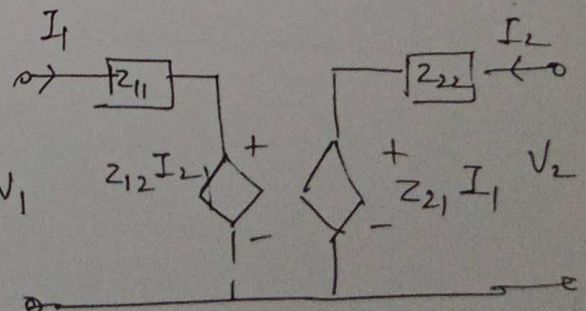
① open circuit Impedance parameters or Z-para.

$$(V_1, V_2) = f(I_1, I_2)$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$



$$\frac{V_2 = 0}{I_1 = I_1}$$

$$I_1 = I_1 \quad V_2 = 0$$

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

i/p driving point impedance

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

forward transfer impedance

$$\frac{I_1 = 0}{V_2 = V_2}$$

$$V_2 = V_2 \quad I_1 = 0$$

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

reverse transfer impedance

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

o/p driving pt. imp.

② Short ckt Admittance or Y-par.

$$(I_1, I_2) = f(V_1, V_2)$$

$$[I] = [Y][V]$$

$$Y = Z^{-1}$$

$$Y_{ij} \neq \frac{1}{Z_{ij}}$$

③ Transmission (T) or chain or ABCD para. ②

$$(V_1, I_1) = f(V_2, -I_2)$$

T-para is used in the analysis of transmission line.
O/p port $\rightarrow$ receiving end
 $\times$ I/p port $\rightarrow$ sending end.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

④ Inverse T parameters

$$(V_2, I_2) = f(V_1, -I_1)$$

$$\begin{bmatrix} V_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} V_1 \\ -I_1 \end{bmatrix}$$

⑤ Hybrid parameters : (h-para.)

widely used in electronic ckt, esp in constructing models for transistors.

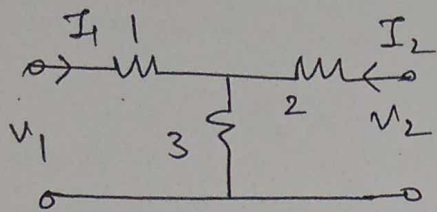
$$(V_1, I_2) = f(I_1, V_2)$$

⑥ Inverse hybrid / g-para.

$$: [g] = [h^{-1}]$$

$$(I_1, V_2) = f(V_1, I_2)$$

Q



Calc. ① Z, Y, T, T' h4f.
-parameters.

loop eq.

$$V_1 = 1 I_1 + 3(I_1 + I_2) = 4I_1 + 3I_2$$

$$V_2 = 2I_2 + 3(I_1 + I_2) = 3I_1 + 5I_2$$

① Z-param $(V_1, V_2) = f(I_1, I_2)$.

(a) $I_2 = 0$.

$$V_1 = 4I_1$$

$$V_2 = 3I_1$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{4I_1}{I_1} = 4\Omega$$

$$Z_{21} = \frac{V_2}{I_1} = 3$$

(b) $I_1 = 0$

$$V_1 = 3I_2 \quad \& \quad V_2 = 5I_2$$

$$Z_{12} = 3 \quad \& \quad Z_{22} = 5$$

② Y-par $(I_1, I_2) = f(V_1, V_2)$.

a) $V_2 = 0$.

$$3I_1 = -5I_2$$

$$V_1 = 4I_1 + 3I_2$$

$$V_1 = 4I_1 + 3\left(-\frac{3}{5}\right)I_1$$

$$V_1 = \left(\frac{20-9}{5}\right)I_1$$

$$V_1 = \frac{11}{5}I_1$$

$$Y_{11} = \frac{5}{11}\Omega$$

$$V_1 = 4\left(-\frac{5}{3}\right)I_2 + 3I_2$$

$$V_1 = \left(\frac{-20+9}{3}\right)I_2$$

$$V_1 = -\frac{11}{3}I_2$$

$$Y_{21} = -\frac{3}{11}\Omega$$

$$\frac{V_1 = 0}{Y_{12} = -\frac{3}{11} \Omega} \quad Y_{22} = \frac{4}{11} \Omega$$

③ T-parameters

$$(V_1, I_1) \text{ as a function of } (V_2, -I_2)$$

① $I_2 = 0$

$$V_1 = 4 I_1$$

$$V_2 = 3 I_1$$

$$A = \left. \frac{V_1}{V_2} \right|_{I_2 = 0} = \frac{4}{3}$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2 = 0} = \frac{1}{3} \Omega$$

② $V_2 = 0$

$$3 I_1 = -5 I_2$$

$$V_1 = 4 \left(-\frac{5}{3} I_2 \right) + 3 I_2 = -\frac{11}{3} I_2$$

$$B = \frac{V_1}{-I_2} = \frac{11}{3} \Omega$$

$$D = \frac{I_1}{-I_2} = \frac{5}{3}$$

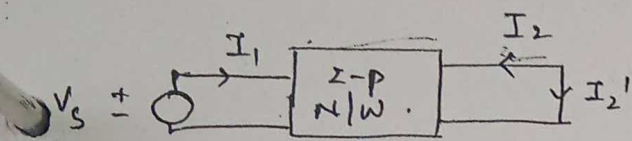
Condition for Reciprocity :

(4)

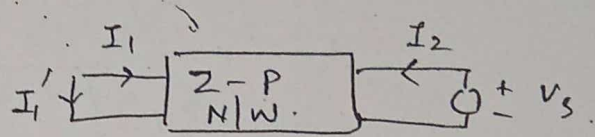
2-port N/w is reciprocal, if the ratio of excitation to response is invariant to an interchange of their positions.

N/w having R, L, C $\rightarrow$ reciprocal

" " dependent sources $\rightarrow$ non-reciprocal.



$$V_1 = V_s, \quad I_1 = I_1 \\ V_2 = 0, \quad I_2 = I_2'$$



$$V_1 = 0, \quad I_1 = I_1' \\ V_2 = V_s, \quad I_2 = I_2'$$

$$\frac{V_s}{I_2'} = \frac{V_s}{I_1'} \quad \text{or} \quad I_2' = I_1'$$

(1) In terms of Z-parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad (a)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad (b)$$

$$V_1 = V_s, \quad I_1 = I_1 \\ V_2 = 0, \quad I_2 = -I_2'$$

$$V_s = Z_{11} I_1 + Z_{12} I_2' \quad \text{--- (1)}$$

$$0 = Z_{21} I_1 + Z_{22} I_2' \quad \text{--- (2)}$$

$$\text{from (2)} \quad I_1 = \frac{-Z_{22} I_2'}{Z_{21}}$$

$$V_s = \left(Z_{11} \frac{-Z_{22} I_2'}{Z_{21}} + Z_{12} I_2' \right) I_2'$$

$$I_2' = \frac{V_s Z_{21}}{Z_{11} Z_{22} - Z_{12} Z_{21}} \quad \text{--- (3)}$$

|| by on putting in a & b.

$$\begin{aligned} V_2 &= V_s & I_2 &= I_2 \\ V_1 &= 0 & I_1 &= -I_1' \end{aligned}$$

we get

$$I_1' = \frac{V_s Z_{12}}{Z_{11} Z_{22} - Z_{12} Z_{21}} \quad \text{--- (4)}$$

On comparing (3) & (4)

$$\boxed{Z_{21} = Z_{12}}$$

(2) In terms of T-parameters

$$Y_{21} = Y_{12}$$

(3) T-parameters

$$V_1 = A V_2 + B(-I_2)$$

$$I_1 = C V_2 + D(-I_2)$$

$$V_1 = V_s \quad I_1 = I_1$$

$$V_2 = 0 \quad I_2 = -I_2'$$

$$V_s = B I_2' \Rightarrow I_2' = \frac{V_s}{B}$$

$$I_1 = D I_2'$$

$$\text{--- (1)}$$

Now

$$V_2 = V_s \quad I_2 = I_2'$$

$$V_1 = 0 \quad I_1 = -I_1'$$

(5)

$$0 = A V_s + B(-I_2) \Rightarrow I_2 = \frac{A V_s}{B}$$

$$-I_1' = C V_s + B(-I_2)$$

$$-I_1' = C V_s + \frac{D A V_s}{B}$$

$$I_1' = \frac{(-BC + DA) V_s}{B} \quad \text{--- (2)}$$

Comparing (1) & (2)

$$\boxed{\begin{array}{l} AD - BC = 1 \\ \text{or } \Delta T = 1 \end{array}}$$

(4) In terms of T' -parameters.

$$\underline{A'D' - B'C'} = 1 \quad \text{or } \underline{\Delta T' = 1}$$

(5) h -parameters

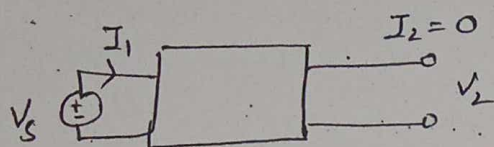
$$\underline{h_{12}} = -h_{21}$$

(6) g -parameters

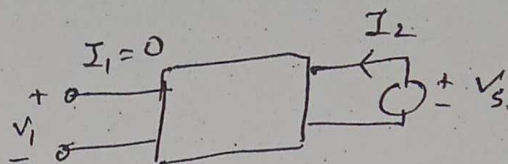
$$\underline{g_{12}} = -g_{21}$$

Condition for Symmetry :

A 2-port N/w. is symmetrical if the ports can be interchanged without changing the port voltages & currents.



$$\begin{aligned} V_1 &= V_s & V_2 &= V_2 \\ I_1 &= I_1 & I_2 &= 0 \end{aligned}$$



$$\begin{aligned} V_1 &= V_1 & I_1 &= 0 \\ V_2 &= V_s & I_2 &= I_2 \end{aligned}$$

$$\left. \frac{V_s}{I_1} \right|_{I_2=0} = \left. \frac{V_s}{I_2} \right|_{I_1=0}$$

(1) In terms of Z-parameters :

$$\cancel{V_1 = V_s}$$

$$\left. \frac{V_s}{I_1} \right|_{I_2=0} = Z_{11}$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$\left. \frac{V_s}{I_2} \right|_{I_1=0} = Z_{22}$$

(2) Y-para.

$$Y_{11} = Y_{22}$$

(3) T-parameters
A = D.

(4). T'-parameters

$$A' = D'$$

(5) h-para

$$h_{11} h_{22} - h_{12} h_{21} = 1$$

$$\text{or } \Delta h = 1$$

(6) g-para

$$\Delta g = 1 \quad \text{or} \quad g_{11} g_{22} - g_{12} g_{21} = 1$$

Relationships b/w parameter Sets:

(i) Z-parameters in terms of other

(a) Z in terms of Y-par.

$$[I] = [Y][V]$$

$$[V] = [Z][I]$$

$$\therefore [Z] = [Y]^{-1}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \frac{1}{\Delta Y} \begin{bmatrix} Y_{22} & -Y_{11} \\ -Y_{21} & Y_{11} \end{bmatrix}$$

$$Z_{11} = \frac{Y_{22}}{\Delta Y} \quad Z_{12} = \frac{-Y_{11}}{\Delta Y} \quad Z_{21} = \frac{-Y_{21}}{\Delta Y}$$

$$Z_{22} = \frac{Y_{11}}{\Delta Y}$$

(b) Z in terms of T-para.

$$V_1 = A V_2 + B(-I_2) \quad - (1)$$

$$I_1 = C V_2 + D(-I_2) \quad - (2)$$

$\Downarrow$

$$V_2 = \frac{1}{C} I_1 + \frac{D}{C} I_2$$

$$\boxed{Z_{21} = \frac{1}{C} ; Z_{22} = \frac{D}{C}}$$

from (1)

$$V_1 = A \left[\frac{1}{C} I_1 + \frac{D}{C} I_2 \right] - B I_2$$

$$V_1 = \frac{A}{C} I_1 + \left(\frac{AD}{C} - B \right) I_2$$

$$\boxed{Z_{11} = \frac{A}{C} \quad Z_{12} = \frac{AD - BC}{C} = \frac{\Delta T}{C}}$$

(c) in T'-para.

$$Z_{21} = \frac{1}{C'}, \quad Z_{22} = \frac{D'}{C'}$$

$$Z_{11} = \frac{A'}{C'} \quad \& \quad Z_{12} = \frac{\Delta T'}{C'}$$

(d) h-parameters :

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$V_2 = \frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1$$

$$\Rightarrow Z_{21} = \frac{-h_{21}}{h_{22}} \quad \& \quad Z_{22} = \frac{1}{h_{22}}$$

$$V_1 = h_{11} I_1 + h_{12} \left(\frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1 \right)$$

$$V_1 = \frac{h_{22} h_{11} - h_{21} h_{12}}{h_{22}} I_1 + \frac{h_{12}}{h_{22}} I_2$$

$$\text{So} \Rightarrow Z_{11} = \frac{\Delta h}{h_{22}} \quad \& \quad Z_{12} = \frac{h_{12}}{h_{22}}$$

(e) g-parameters :

$$Z_{11} = \frac{1}{g_{11}} \quad ; \quad Z_{12} = \frac{-g_{12}}{g_{11}} \quad ; \quad Z_{21} = \frac{-g_{21}}{g_{11}}$$

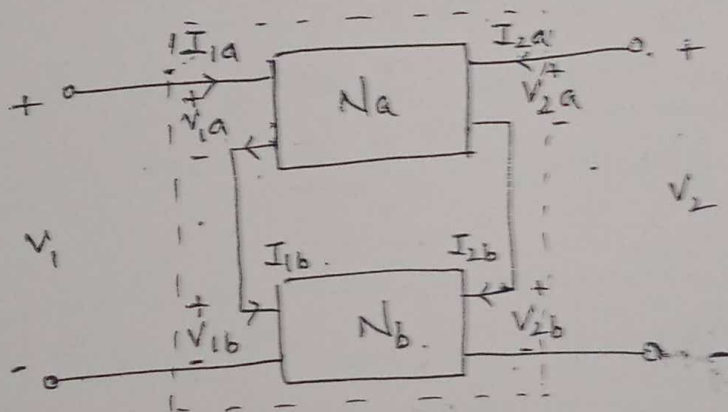
$$Z_{22} = \frac{g_{22}}{g_{11}} \quad \checkmark$$

Interconnection of two-Port N/w.

2-port N/w can be connected as series, parallel, cascade, series-parallel & parallel-series connection.

For each configuration certain set of parameters may be more useful than others to describe the n/w.

① Series Connection:



Na

$$\begin{bmatrix} V_{1a} \\ V_{2a} \end{bmatrix} = \begin{bmatrix} Z_{11a} & Z_{12a} \\ Z_{21a} & Z_{22a} \end{bmatrix} \begin{bmatrix} I_{1a} \\ I_{2a} \end{bmatrix}$$

Nb

$$\begin{bmatrix} V_{1b} \\ V_{2b} \end{bmatrix} = \begin{bmatrix} Z_{11b} & Z_{12b} \\ Z_{21b} & Z_{22b} \end{bmatrix} \begin{bmatrix} I_{1b} \\ I_{2b} \end{bmatrix}$$

for Series Connection

$$I_1 = I_{1a} = I_{1b}, \quad I_2 = I_{2a} = I_{2b}.$$

$$V_1 = V_{1a} + V_{1b} \quad V_2 = V_{2a} + V_{2b}.$$

$$V_1 = V_{1a} + V_{1b} = Z_{11a} I_{1a} + Z_{12a} I_{2a} + Z_{11b} I_{1b} + Z_{12b} I_{2b}.$$

$$V_1 = (Z_{11a} + Z_{11b}) I_1 + (Z_{12a} + Z_{12b}) I_2.$$

$$\text{Similarly } V_2 = V_{2a} + V_{2b} = (Z_{21a} + Z_{21b}) I_1 + (Z_{22a} + Z_{22b}) I_2.$$

$$Z_{11} = Z_{11a} + Z_{11b}.$$

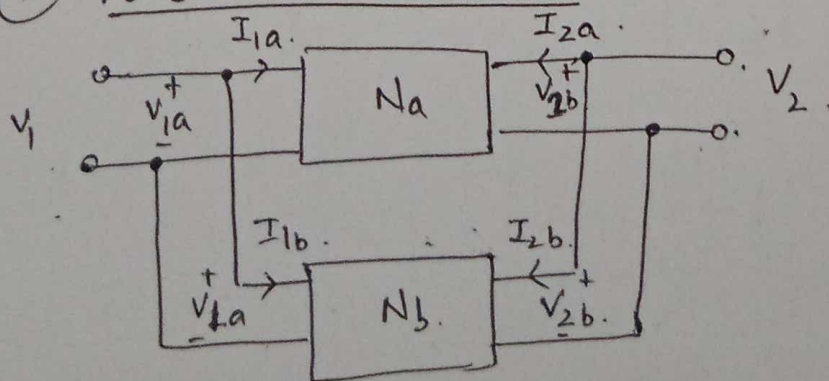
$$Z_{12} = Z_{12a} + Z_{12b}.$$

$$Z_{21} = Z_{21a} + Z_{22b}.$$

$$Z_{22} = Z_{22a} + Z_{22b}.$$

$$\text{So } [Z] = [Z_a] + [Z_b]$$

② Parallel connection:



$$\frac{N/a}{\begin{bmatrix} I_{1a} \\ I_{2a} \end{bmatrix}} = \begin{bmatrix} Y_{11a} & Y_{12a} \\ Y_{21a} & Y_{22a} \end{bmatrix} \begin{bmatrix} V_{1a} \\ V_{2a} \end{bmatrix}$$

$$\frac{N/b}{\begin{bmatrix} I_{1b} \\ I_{2b} \end{bmatrix}} = \begin{bmatrix} Y_{11b} & Y_{12b} \\ Y_{21b} & Y_{22b} \end{bmatrix} \begin{bmatrix} V_{1b} \\ V_{2b} \end{bmatrix}$$

for || connection $\rightarrow$

$$V_1 = V_{1a} = V_{1b} \quad V_2 = V_{2a} = V_{2b}$$

$$I_1 = I_{1a} + I_{1b} \text{ --- (1)} \quad I_2 = I_{2a} + I_{2b} \text{ --- (2)}$$

in (1).

$$I_1 = Y_{11a} V_{1a} + Y_{12a} V_{2a} + Y_{11b} V_{1b} + Y_{12b} V_{2b}$$

$$I_1 = (Y_{11a} + Y_{11b}) V_{1a} + (Y_{12a} + Y_{12b}) V_{2a}$$

$$I_2 = (Y_{21a} + Y_{21b}) V_1 + (Y_{22a} + Y_{22b}) V_2$$

$$\text{So } [Y] = [Y_a] + [Y_b]$$

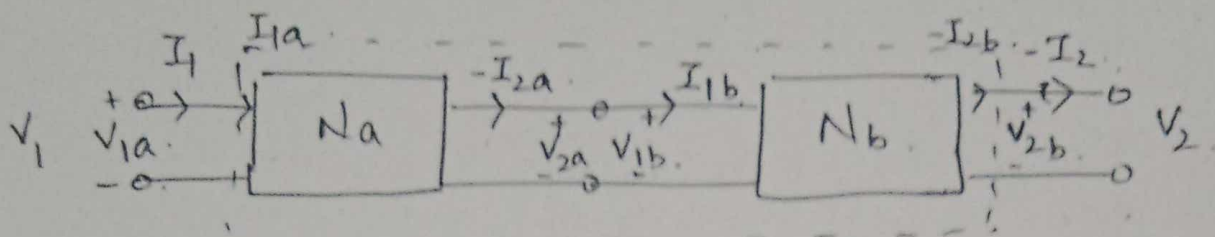
$$Y_{11} = Y_{11a} + Y_{11b} \quad Y_{22} = Y_{22a} + Y_{22b}$$

$$Y_{12} = Y_{12a} + Y_{12b}$$

$$Y_{21} = Y_{21a} + Y_{21b}$$

③ Cascade Connection: (Tandem)

⑨



This is the simplest possible interconnection
2 - N/w r cascaded when O/P of 1st N/w
is the I/P of the 2nd N/w.

$$\underline{N_a} \begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix}$$

$$\underline{N_b} \begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix} = \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

In Cascade

$$V_1 = V_{1a} \quad ; \quad V_{2a} = V_{1b}$$

$$V_2 = V_{2b}$$

$$I_1 = I_{1a} \quad -I_{2a} = I_{1b} \quad I_{2b} = I_2$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix} =$$

$$= \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{1b} \\ -I_{1b} \end{bmatrix}$$

$$= \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\text{So } \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix}$$

$$A = A_a A_b + B_a C_b, \quad C = C_a A_b + D_a C_b$$

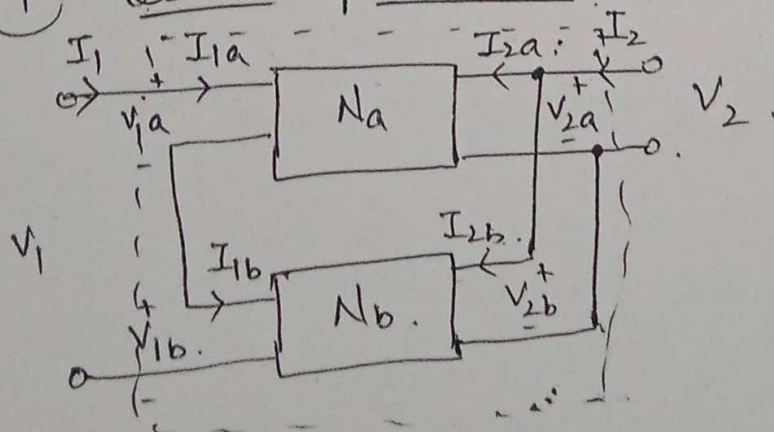
$$B = B_a B_b + B_a D_b, \quad D = C_a B_b + D_a D_b$$

$$[T] = [T_a] \cdot [T_b]$$

In terms of T'-parameters

$$[T'] = [T'_b] \cdot [T'_a]$$

④ Series-parallel



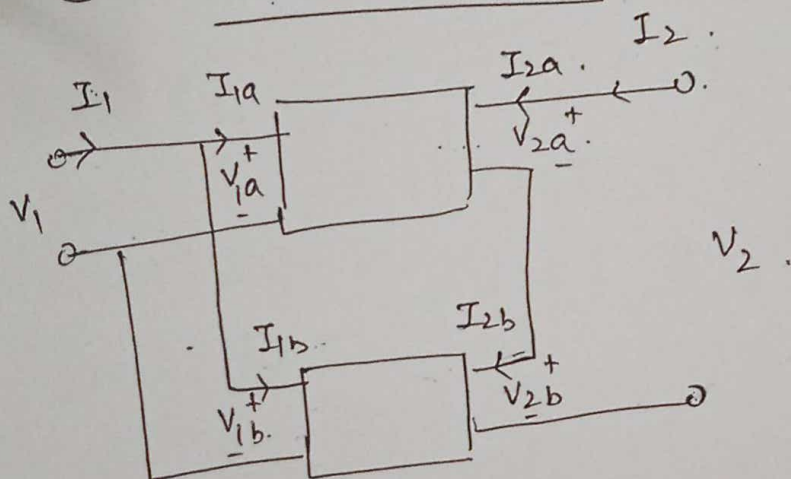
for Series-parallel.

$$V_1 = V_{1a} + V_{1b} \quad V_2 = V_{2a} = V_{2b}$$

$$I_1 = I_{1a} = I_{1b} \quad I_2 = I_{2a} + I_{2b}$$

here $[h] = [h_a] + [h_b]$.

⑤ Parallel-Series



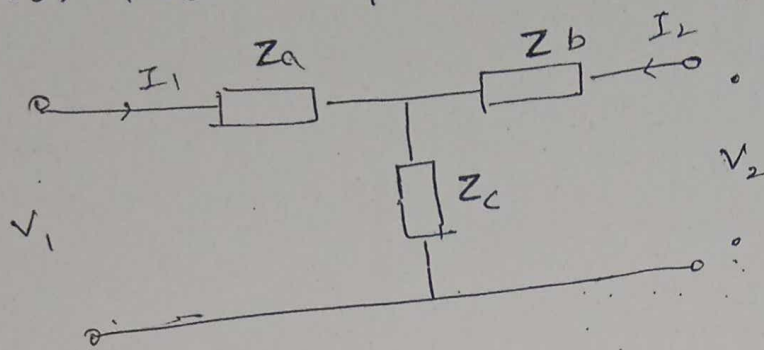
$$V_1 = V_{1a} = V_{1b} \quad I_1 = I_{1a} + I_{1b}$$

$$V_2 = V_{2a} + V_{2b} \quad I_2 = I_{2a} = I_{2b}$$

here $[g] = [g_a] + [g_b]$.

Ques:

① For the T-n/w shown find Z-parameters.



$$V_1 = Z_a I_1 + Z_c (I_1 + I_2)$$

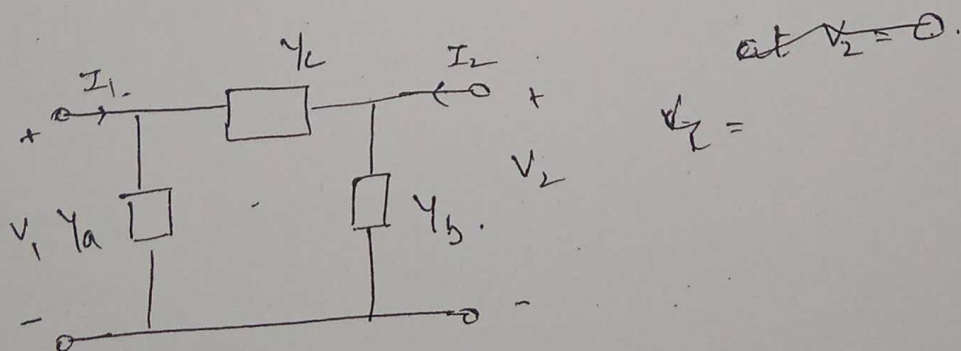
$$= (Z_a + Z_c) I_1 + Z_c I_2$$

$$V_2 = Z_b I_2 + Z_c (I_1 + I_2)$$

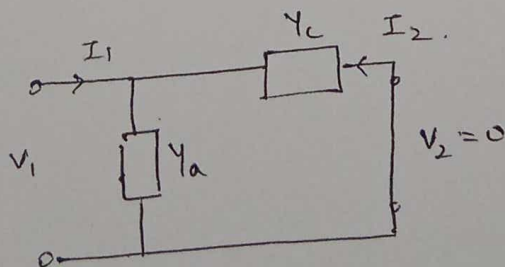
$$= Z_c I_1 + (Z_b + Z_c) I_2$$

$$Z = \begin{bmatrix} Z_a + Z_c & Z_c \\ Z_c & Z_b + Z_c \end{bmatrix}$$

②



at $V_2 = 0$



$$\left. \begin{aligned} V_1 &= \frac{(I_1 + I_2)}{Y_a} \\ V_1 &= -\frac{I_2}{Y_c} \end{aligned} \right\} \text{①}$$

at $V_1 = 0$

$$V_2 = \frac{(I_1 + I_2)}{Y_b}$$

$$\& V_2 = -\frac{I_1}{Y_c} \rightarrow \text{②}$$

from (1)

$$I_2 = (-Y_c) V_1$$

$$V_1 = [I_1 + V_1(-Y_c)] \frac{1}{Y_a}$$

$$(Y_a + Y_c) V_1 = I_1$$

$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} = Y_a + Y_c$$

$$Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} = -Y_c$$

from (2) &
 $V_1 = 0$

$$V_2 = (I_1 + I_2) \frac{1}{Y_b}$$

$$V_2 = -\frac{I_1}{Y_c}$$

$$\Rightarrow Y_{12} = \frac{I_1}{V_2} = -Y_c$$

$$V_2 = (-Y_c V_2 + I_2) \frac{1}{Y_b}$$

$$Y_b V_2 + Y_c V_2 = I_2$$

$$(Y_b + Y_c) V_2 = I_2$$

$$Y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0} = Y_b + Y_c$$

$$Y = \begin{bmatrix} Y_a + Y_c & -Y_c \\ -Y_c & Y_b + Y_c \end{bmatrix}$$

Open ckt & short ckt impedances

open ckt impedances $\rightarrow$ measurement

$$Z_{10} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \equiv Z_{11}$$

at o/p port while o/p $\rightarrow$ is open ckt

$$Z_{20} = \left. \frac{V_2}{I_2} \right|_{I_1=0} \equiv Z_{22}$$

short ckt impedances $\rightarrow$

$$Z_{1s} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \equiv \frac{1}{Y_{11}}$$

$$Z_{2s} = \left. \frac{V_2}{I_2} \right|_{V_1=0} \equiv \frac{1}{Y_{22}}$$

In terms of T-parameters

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

$$Z_{10} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = \frac{A}{C}$$

$$Z_{20} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = \frac{D}{C}$$

$$Z_{1s} = \left. \frac{V_1}{I_1} \right|_{V_2=0} = \frac{B}{D}$$

$$Z_{2s} = \left. \frac{V_2}{I_2} \right|_{V_1=0} = \frac{B}{A}$$

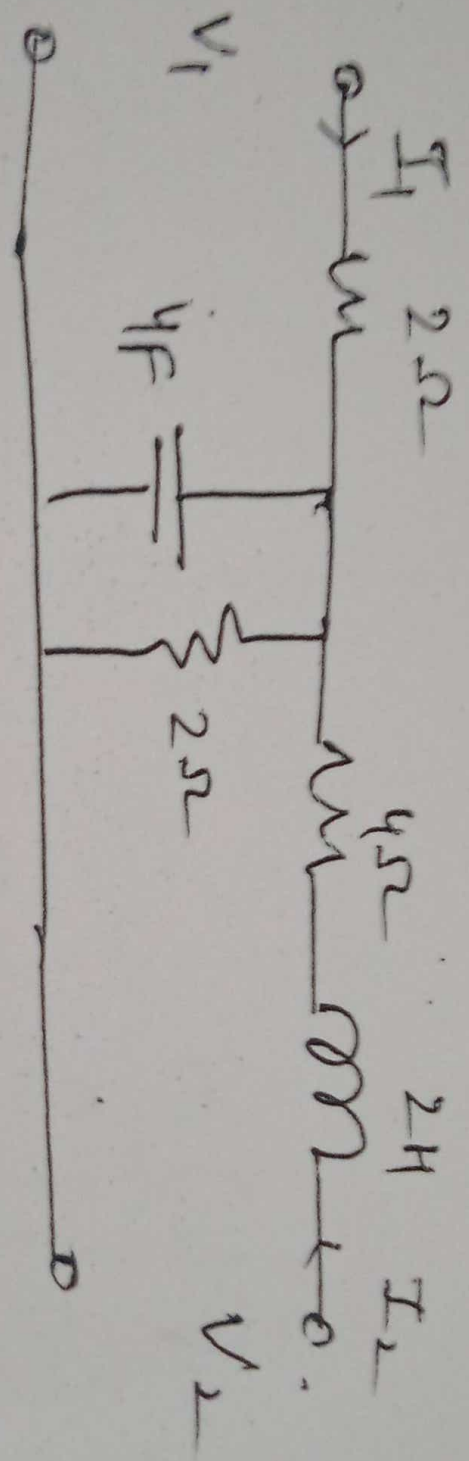
T-para. in terms of
o.c & s.c. Impedance

$$A = \pm \sqrt{\frac{Z_{10}}{Z_{20} - Z_{2s}}}$$

$$B = \pm Z_{2s} \sqrt{\frac{Z_{10}}{Z_{20} - Z_{2s}}}$$

$$C = \pm \sqrt{\frac{1}{Z_{10}(Z_{20} - Z_{2s})}}$$

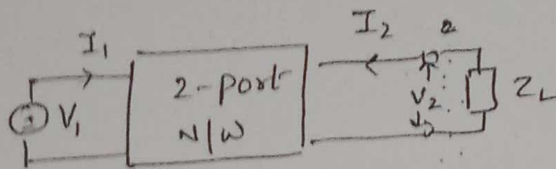
$$D = \pm \frac{Z_{20}}{\sqrt{Z_{10}(Z_{20} - Z_{2s})}}$$



Input impedance

If load imp. Z_L is connected to O/P port.

then $Z_{ip} = \frac{V_1}{I_1}$ is I/P Z.



① Z_{ip} in terms of Z

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$V_2 = -I_2 Z_L$$

$$-I_2 Z_L = Z_{21} I_1 + Z_{22} I_2$$

$$I_2 = \frac{-Z_{21}}{Z_{22} + Z_L} I_1$$

$$V_1 = Z_{11} I_1 + Z_{12} \left[\frac{-Z_{21}}{Z_{22} + Z_L} \right] I_1$$

$$Z_{ip} = \frac{V_1}{I_1} = \frac{Z_{11} Z_{22} - Z_{21} Z_{12} + Z_{11} Z_L}{Z_{22} + Z_L}$$

② in terms of T-para.

$$Z_{ip} = \frac{V_1}{I_1} = \frac{A Z_L + B}{C Z_L + D}$$

I/p Imp. in terms of h-parameters

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$V_2 = -I_2 Z_L$$

$$V_2 = -Z_L (h_{21} I_1 + h_{22} V_2)$$

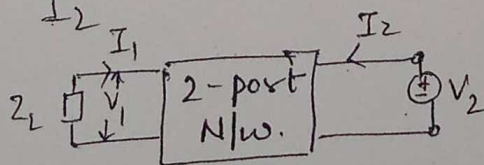
$$V_2 = \frac{-h_{21} Z_L}{1 + h_{22} Z_L} I_1$$

$$V_1 = h_{11} I_1 + h_{12} \left[\frac{-h_{21} Z_L}{1 + h_{22} Z_L} \right] I_1$$

$$Z_{ip} = \frac{V_1}{I_1} = h_{11} - \frac{h_{12} h_{21} Z_L}{1 + h_{22} Z_L}$$

O/p Impedance

If a load Imp. Z_L is connected to o/p port then $Z_{op} = \frac{V_2}{I_2}$ is termed as o/p Imp.



① O/p Imp. in terms of Z-parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$V_1 = -I_1 Z_L$$

$$I_1 = - \frac{(Z_{11} I_1 + Z_{12} I_2)}{Z_L}$$

$$I_1 = - \frac{Z_{12}}{Z_{11} + Z_L} I_2$$

$$V_2 = Z_{21} \left[- \frac{Z_{12}}{Z_{11} + Z_L} \right] I_2 + Z_{22} I_2$$

$$Z_{op} = \frac{V_2}{I_2} = \frac{Z_{11} Z_{22} - Z_{21} Z_{12} + Z_{22} Z_L}{Z_{11} + Z_L}$$

② in terms of T-par.

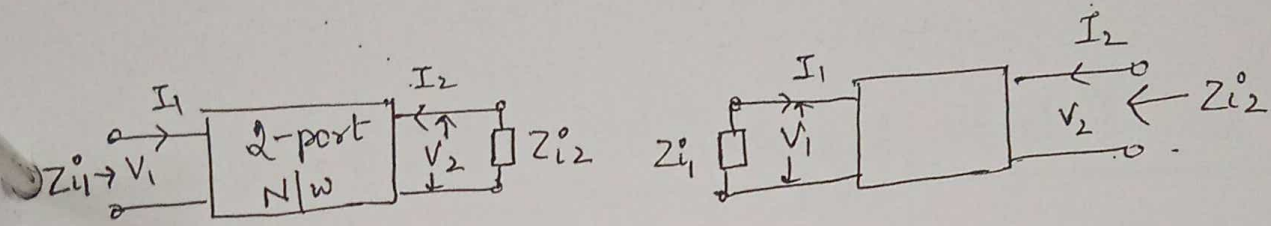
$$Z_{op} = \frac{D Z_L + B}{C Z_L + A}$$

③ h-par

$$Z_{op} = \frac{h_{11} + Z_L}{h_{11} h_{22} - h_{12} h_{21} + h_{22} Z_L}$$

Image Impedances :

In a 2-port N/w, if the impedance at I/P port with impedance Z_{i2} connected across O/P port be Z_{i1} & the imp. at O/P port with imp Z_{i1} connected across I/P port be Z_{i2} , then Z_{i1} & Z_{i2} r image impedance



For symmetrical n/w

$Z_{i1} = Z_{i2}$ & is called characteristic or iterative imp. Z_c .

$Z_{i1} = \frac{V_1}{I_1}$ (driving point imp. at I/P port)

$Z_{i2} = \frac{V_2}{I_2}$ (" " " " O/P ")

Image Imp. in terms of I/P & O/P Imp.

$Z_{i1} = Z_{ip}$ & $Z_{i2} = Z_{op}$

in terms of T-parameters

$Z_{i1} = Z_{ip}$

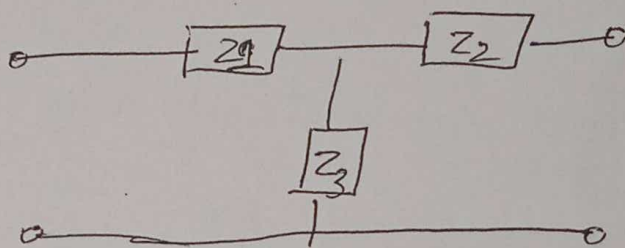
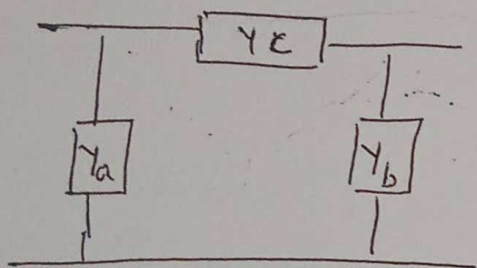
Solving (1) & (2).
 $Z_{i1} = \sqrt{\frac{AB}{CD}}$

in terms of o/p ukt & s.c. imp:

$$Z_{i1} = \sqrt{\frac{AB}{CD}} = \sqrt{\frac{A}{C} \cdot \frac{B}{D}} = \sqrt{Z_{10} \cdot Z_{15}}$$

$$Z_{i2} = \sqrt{\frac{BD}{AC}} = \sqrt{\frac{D}{C} \cdot \frac{B}{A}} = \sqrt{Z_{20} \cdot Z_{25}}$$

Π to T transformation

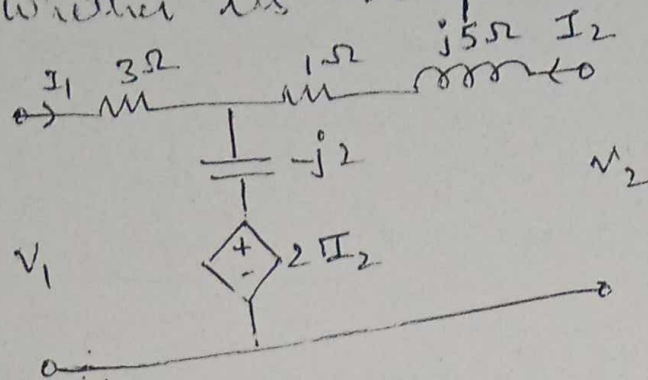


Π to T

$$Z_1 = \frac{\frac{1}{Y_a} \cdot \frac{1}{Y_c}}{\frac{1}{Y_a} + \frac{1}{Y_b} + \frac{1}{Y_c}} = \frac{\frac{1}{Y_a Y_c}}{\frac{Y_b Y_c + Y_c Y_a + Y_a Y_b}{Y_a Y_b Y_c}} = \frac{Y_b}{Y_a Y_b + Y_b Y_c + Y_c Y_a}$$

T to Π

15
Ques Calc. z-par. of the N/w shown & find out whether its reciprocal or symmetrical or none.



$$V_1 = 3I_1 - j2(I_1 + I_2) + 2I_2$$

$$V_1 = (3 - j2)I_1 + 2I_2(1 - j)$$

$$V_2 = I_2 + j5I_2 - j2(I_1 + I_2) + 2I_2$$

$$V_2 = (3 + 3j)I_2 - j2I_1$$

$$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0} = (3 - j2)\Omega$$

$$Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0} = -j2\Omega$$

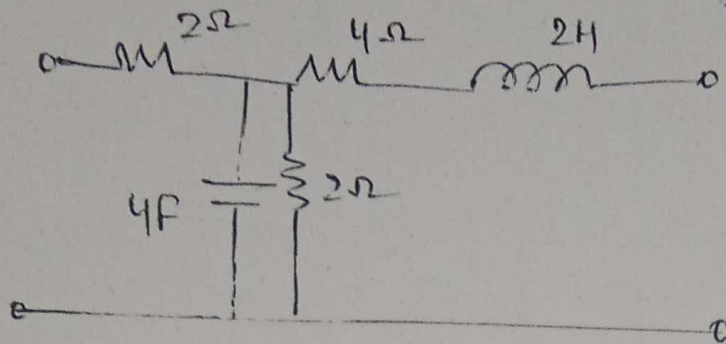
$$Z_{12} = (2 - j2)\Omega$$

$$Z_{22} = (3 + 3j)\Omega$$

Since $Z_{12} \neq Z_{21}$ so not reciprocal

$Z_{11} \neq Z_{22}$ so not symm.

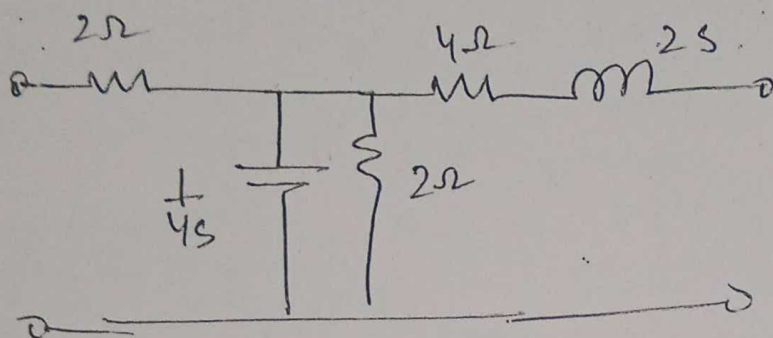
Q Obtain 2-parameters of the N/w.



$$Z = R \pm Ls \equiv \frac{1}{Cs}$$

$$Y \equiv \frac{1}{R} \pm \frac{1}{Ls} \equiv Cs$$

Converting into s-domain.



$$\frac{1}{4s} \parallel 2\Omega = \frac{2}{1+8s}$$

$$Z_a = 2\Omega \quad Z_b = 4+2s$$

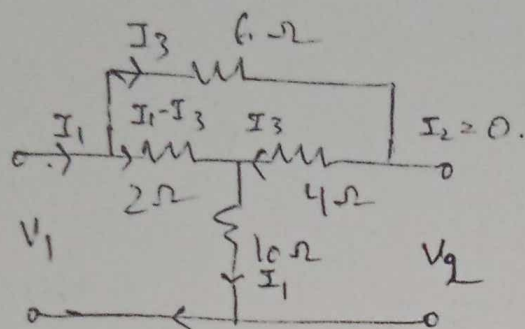
$$Z_c = \frac{2}{1+8s}$$

$$\text{So } Z_{11} = Z_a + Z_c = 2 + \frac{2}{1+8s} = \frac{4+16s}{1+8s}$$

$$Z_{12} = Z_{21} = Z_c = \frac{2}{1+8s}$$

$$Z_{22} = Z_b + Z_c = 4+2s + \frac{2}{1+8s} = \frac{6+34s+16s^2}{1+8s}$$

Q. obtain the open circuit param. of the N/w.



$$I_2 = 0$$

$$V_1 = 2(I_1 - I_3) + 10 I_1$$

$$V_1 = 12 I_1 - 2 I_3$$

$$V_2 = 4 I_3 + 10 I_1$$

$$I_3 = I_1 \times \frac{2}{2+6+4}$$

$$= \frac{2 I_1}{12} = \frac{I_1}{6}$$

$$V_1 = 2 \left(I_1 - \frac{I_1}{6} \right) + 10 I_1$$

$$= \frac{10}{6} I_1 + 10 I_1$$

$$= \frac{70}{6} I_1 = \frac{35}{3} I_1$$

$$V_2 = 4 \frac{I_1}{6} + 10 I_1 = \frac{32}{3} I_1$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{35}{3} \Omega$$

$$Z_{21} = \frac{V_2}{I_1} = \frac{32}{3} \Omega$$

for $I_1 = 0$

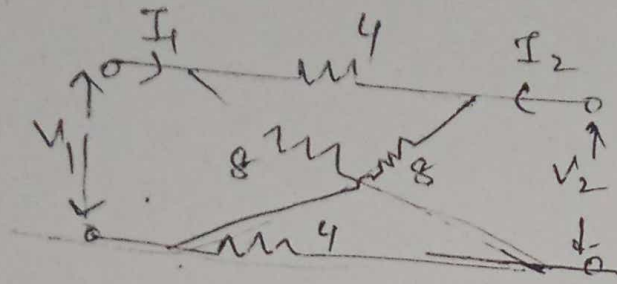
$$V_1 = 2 I_4 + 10 I_2$$

$$V_2 = 4(I_2 - I_4) + 10 I_2$$

$$Z_{12} = \frac{V_1}{I_2} = \frac{32}{3} \Omega$$

$$Z_{22} = \frac{V_2}{I_2} = \frac{38}{3} \Omega$$

Q For the lattice N/w find 2-parameter



Case 1 $I_2 = 0$

$$V_1 = 4 \frac{I_1}{2} + 8 \frac{I_1}{2} = 6 I_1$$

$$V_2 = 8 \frac{I_1}{2} - 4 \frac{I_1}{2} = 2 I_1$$

Case 2 $I_1 = 0$

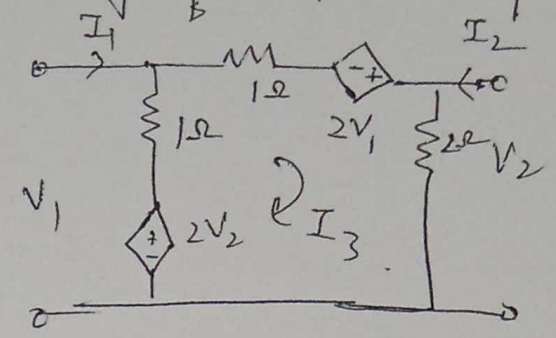
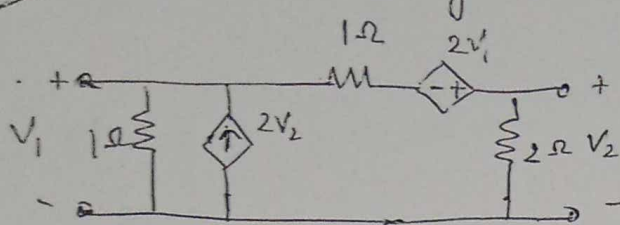
$$V_1 = 8 \frac{I_2}{2} - 4 \frac{I_2}{2} = 2 I_2$$

$$V_2 = 6 I_2$$

$$Z_{11} = Z_{22} = 6 \Omega$$

$$Z_{12} = Z_{21} = 2 \Omega$$

Q. For the given circuit find V_1 & V_2 per.



Solⁿ

$$V_1 = 1(I_1 - I_3) + 2V_2$$

$$V_1 = I_1 - I_3 + 2V_2$$

$$I_3 = I_1 + 2V_2 - V_1$$

$$\rightarrow 1 \cdot I_3 - 2V_1 + 2(I_2 + I_3) - 2V_2 + 1(I_3 - I_1) = 0$$

$$4I_3 + 2I_2 - I_1 = 2V_1 + 2V_2 \quad \text{--- (1)}$$

$$\rightarrow V_2 = 2(I_2 + I_3) \quad \text{--- (2)}$$

Putting I_3 in (1)

$$2V_1 + 2V_2 = 4(I_1 + 2V_2 - V_1) + 2I_2 - I_1$$

$$0 = 3I_1 + 6V_2 - 6V_1 + 2I_2$$

$$6(V_1 - V_2) = 3I_1 + 2I_2 \quad \text{--- (3)}$$

I_3 in (2)

$$V_2 = 2I_2 + 2(I_1 + 2V_2 - V_1)$$

$$0 = 2I_2 + 2I_1 + 3V_2 - 2V_1$$

$$2V_1 - 3V_2 = 2(I_2 + I_1) \quad \text{--- (4)}$$

(3) - (4)

$$4V_1 - 3V_2 = I_1 \quad \text{--- (5)}$$

$$\text{So } I_1 = 4V_1 - 3V_2 \quad \text{--- (a)}$$

Putting (5) in (4).

$$2V_1 - 3V_2 = 2(4V_1 - 3V_2) + 2I_2$$

$$I_2 = -\frac{6}{2}V_1 + \frac{3}{2}V_2$$

$$= -3V_1 + \frac{3}{2}V_2$$

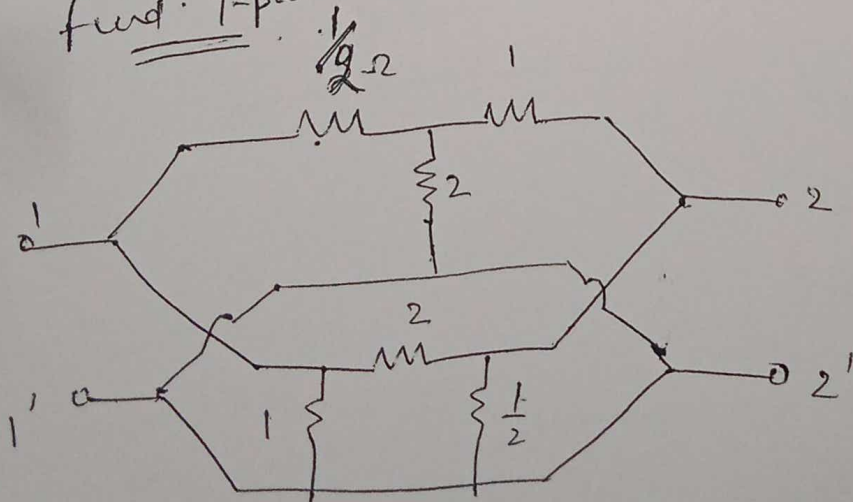
Using (a) & (b)

$$Y = \begin{bmatrix} 4 & -3 \\ -3 & \frac{3}{2} \end{bmatrix}$$

$$Z = [Y]^{-1} = \begin{bmatrix} -\frac{1}{2} & -1 \\ -1 & -\frac{4}{3} \end{bmatrix}$$

find Y-par.

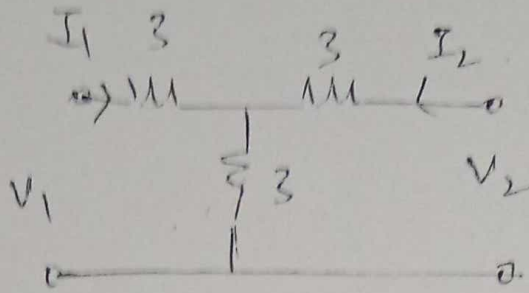
Ques



$$Y_A = [Z_A]^{-1} = \begin{bmatrix} \frac{1}{2} + 2 & 2 \\ 2 & 2 + 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{6}{7} & -\frac{4}{7} \\ -\frac{4}{7} & \frac{5}{7} \end{bmatrix}$$

$$Y_B = \begin{bmatrix} \frac{1}{1} + \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{1} + 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{2} \end{bmatrix} \quad Y = Y_A + Y_B$$

For the given N/w find image para.



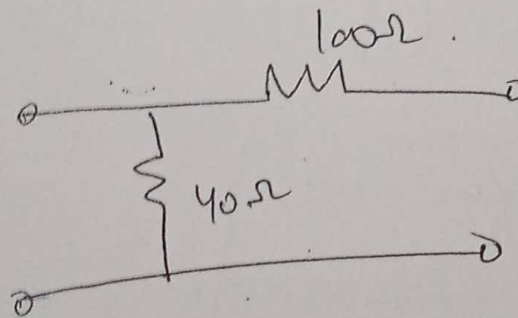
$$A = 2 \begin{bmatrix} \frac{V_1}{V_2} \\ \frac{I_1}{I_2} \end{bmatrix}$$

$$B = 9 \Omega \frac{V_1}{I_2}$$

$$C = \frac{1}{3} \frac{I_1}{V_2}$$

$$D = 2 \begin{bmatrix} \frac{I_1}{I_2} \end{bmatrix}$$

$$Z_{in} = \sqrt{\frac{AB}{CD}} = \sqrt{\frac{2 \times 9}{\frac{1}{3} \times 2}} = \sqrt{27} = 5.2 \Omega$$



h-param.

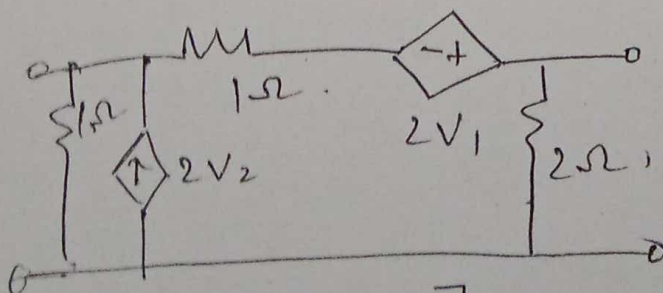
$$h_{11} = \frac{200}{7}$$

$$h_{21} = \frac{-2}{7}$$

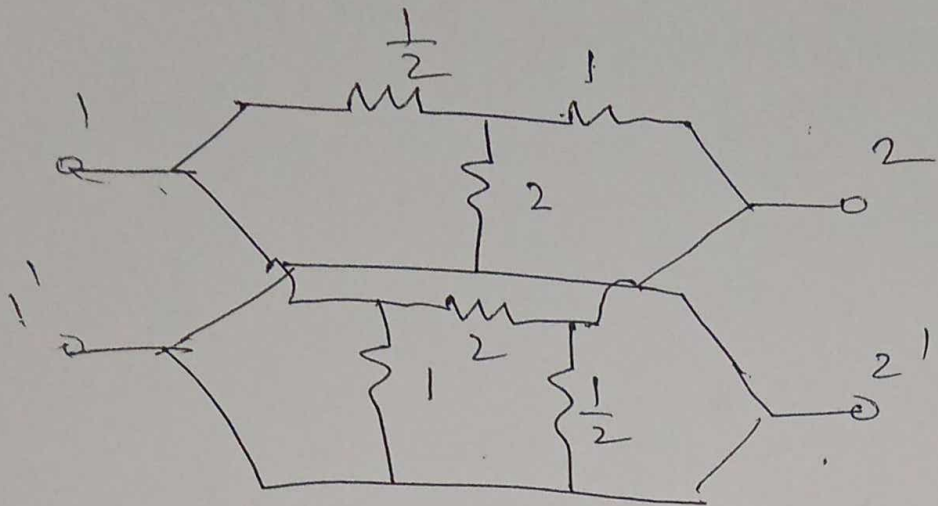
$$h_{12} = \frac{2}{7}$$

$$h_{22} = \frac{1}{140}$$

Y & Z - parameters

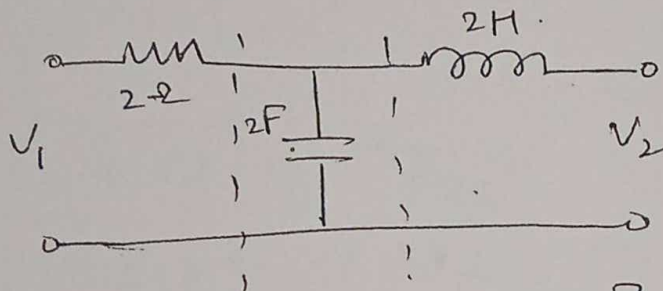


$$Y = \begin{bmatrix} 4 & -3 \\ -3 & \frac{3}{2} \end{bmatrix} \quad Z = \begin{bmatrix} -\frac{1}{2} & -1 \\ -1 & -\frac{4}{3} \end{bmatrix}$$



Y-parameters.

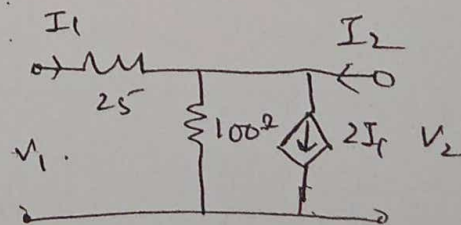
8 Determine T-parameters.



$$T_a = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad T_b = \begin{bmatrix} 1 & 0 \\ 2s & 1 \end{bmatrix} \quad T_c = \begin{bmatrix} 1 & 2s \\ 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 4s+1 & 2(4s^2+s+1) \\ 2s & 4s^2+1 \end{bmatrix}$$

9 Find Y-parameters



$$I_1 + I_2 = \frac{V_2}{100} + 2I_1$$

$$I_1 = -\frac{V_2}{100} + I_2 \quad \text{--- (1)}$$

$$\text{Also } I_1 = \frac{V_1 - V_2}{25} \quad \text{--- (2)}$$

$$I_1 = \frac{1}{25} V_1 - \frac{1}{25} V_2 \quad \text{--- (3)}$$

from (1) & (2).

$$\frac{V_1 - V_2}{25} = -\frac{V_2}{100} + I_2$$

$$I_2 = \frac{1}{25} V_1 - \frac{3}{100} V_2$$

$$Y_{21} = Y_{11} = \frac{1}{25} \quad Y_{12} = \frac{1}{25} \\ Y_{22} = -\frac{3}{100}$$

6. M/w Synthesis

elements of realizability theory:

$$\textcircled{1} \rightarrow T(s) = \frac{R(s)}{E(s)} \quad \begin{matrix} \text{Response} \\ \text{(Excitation)} \end{matrix}$$

① Causality & stability.

↓
response of a n/w ^{must be} = 0 for $t < 0$.

for n/w to be
stable $\rightarrow$ following 3 condⁿ must be
satisfied.

- ① $T(s)$ can't have poles in rgh half of s-pla
- ② $T(s)$ can't have multiple poles in the $(j\omega)$ axis.
- ③ degree of $N(s)$ of $T(s)$ can't exceed the degree of $d(s)$ by more than unity.

$$T(s) = \frac{N(s)}{D(s)}$$

② Hurwitz Polynomial

It is the $D(s)$ polynomial of the n/w funcⁿ satisfying certain condⁿ.

A polynomial $P(s)$ is Hurwitz if.

- ① $P(s)$ is real when s is real.
- ② roots of $P(s)$ have real parts which are 0 or negative.

Properties of Hurwitz

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$$

① All coefficients a_i must be +ve.

⇒ none of the coeff. b/w highest order term in s & the lowest order term may be 0, unless the polynomial is even or odd.

② Both odd & even parts of a Hurwitz polynomial $P(s)$ have roots on $j\omega$ axis only.

even $\rightarrow M(s)$

odd $\rightarrow N(s)$

$$\text{so } P(s) = M(s) + N(s).$$

then $M(s)$ & $N(s)$ have roots on $j\omega$ axis only.

③ If $P(s) \rightarrow$ either even or odd, all its roots are on $j\omega$ axis.

④ Continued fraction expansion of the ratio odd to even or vice versa, yields all the quotient terms:

$$\text{As } \Psi(s) = \frac{N(s)}{M(s)} \text{ or } \frac{M(s)}{N(s)}.$$

$$\Psi(s) = q_1(s) + \frac{1}{q_2(s) + \frac{1}{q_3(s) + \frac{1}{\dots}}}$$

- ⑤ If $P(s) \rightarrow$ Hurwitz poly.
& $w(s) \rightarrow$ multiplicative factor.

Then $P_1(s) = P(s) \cdot w(s)$ is also Hurwitz.
if $w(s)$ is Hurwitz polynomial.

- ⑥ If $P(s) \rightarrow$ only odd or even then.
Cont. fraction expansion is not poss.

So $P(s) \rightarrow$ Hurwitz if $\frac{P(s)}{P'(s)}$ gives
a continued fr. ex.

Q Check whether the given polynomial
 $P(s) = s^4 + s^3 + 5s^2 + 3s + 4$ is Hurwitz or not.

Soln ① all coeff of $P(s)$ are +ve. so $P(s) \rightarrow$ real
for s real.

② even & odd
 $M(s) = s^4 + 5s^2 + 4$
 $N(s) = s^3 + 3s$

$$\begin{aligned} \psi(s) = \frac{M(s)}{N(s)} &= \frac{s^4 + 5s^2 + 4}{s^3 + 3s} \left(\begin{array}{r} s^4 + 5s^2 + 4 \\ s^4 + 3s^2 \\ \hline 2s^2 + 4 \end{array} \right) \left(\begin{array}{r} s^3 + 3s \\ s^3 + 2s \\ \hline s \end{array} \right) \left(\begin{array}{r} 2s^2 + 4 \\ 2s^2 \\ \hline 4 \end{array} \right) \left(\begin{array}{r} 4s \\ 4s \\ \hline s \end{array} \right) \left(\begin{array}{r} s \\ s \\ \hline 0 \end{array} \right) \end{aligned}$$

Q $P(s) = s^4 + s^3 + 2s^2 + 4s + 1$

Solⁿ

① is satisfied.

② $M(s) = s^4 + 2s^2 + 1$

$N(s) = s^3 + 4s$

$\Psi(s) = \frac{M(s)}{N(s)}$

$(s^3 + 4s) \overline{s^4 + 2s^2 + 1} \begin{array}{l} s \\ s^4 + 4s^2 \end{array}$

$\underline{-2s^2 + 1} \quad) \quad s^3 + 4s \quad \left(-\frac{1}{2}s \leftarrow \right.$
 $\quad \quad \quad s^3 - \frac{s}{2}$

$\quad \quad \quad \underline{\frac{9}{2}s} \quad) \quad -2s^2 + 1 \quad \left(+\frac{2}{9} \cdot 2s = -\frac{4}{9}s \leftarrow \right.$
 $\quad \quad \quad \quad \quad \quad -2s^2$

not herwitz.

coz of -ve quotient
term s

$1 \quad) \quad \frac{9}{2}s \quad \left| \quad \frac{9}{2}s \right.$

$\quad \quad \quad \frac{9}{2}s$
 $\quad \quad \quad \underline{\varphi}$

Q Find the range of values of a so that

Solⁿ $P(s) = s^4 + s^3 + as^2 + 2s + 3$ is herwitz.

Guidⁿ ① All a_i must be +ve so $a > 0$.

② $M(s) / N(s)$.

$a > 2$ ~~so~~ $a > 3.5$
 so $a > 3.5$